Yes, water hyacinth ash (WHA) can be used as a partial cement replacement, and it safely locks away the absorbed heavy metals within the concrete matrix through a process called **solidification and stabilization**. Because it exhibits pozzolanic activity, burning the invasive plant under controlled conditions transforms it into a viable supplementary cementitious material. [1, 2, 3, 4]

The scientific breakdown of how this works and how the heavy metals are managed includes:

1. How It Acts Like Cement (Pozzolanic Activity)

Water hyacinth ash is rich in chemical compounds like **silica (SiO_2), calcium oxide (CaO), and aluminum oxide (Al_2O_3)**. When mixed with ordinary Portland cement and water, these oxides react with the calcium hydroxide byproduct generated during cement hydration. This reaction creates extra calcium silicate hydrate (C-S-H) gel, which is the primary compound responsible for concrete's strength and durability. Studies show that replacing **10% to 14.3% of cement** with finely ground WHA can actually improve the compressive, flexural, and tensile strength of the concrete. [1, 5, 6, 7, 8]

2. Trapping the Absorbed Heavy Metals

Water hyacinths are widely used in phytoremediation to absorb toxic heavy metals like lead (Pb), cadmium (Cd), chromium (Cr), and copper (Cu) from polluted water bodies. When the plant is harvested and incinerated, these heavy metals remain concentrated in the ash. However, using this ash in concrete does not present an immediate environmental risk because of how cement hydrates: [5, 9, 10, 11]

- **Chemical Binding:** During the curing process, heavy metal ions are chemically integrated into the crystalline structures of the cement hydration products (like the C-S-H gel matrix). [12]
- **Physical Encapsulation:** The dense, hardened concrete matrix physically traps the remaining metal particles, blocking them from escaping. [12, 13]
- **Leaching Prevention:** Environmental leaching tests (such as the Toxicity Characteristic Leaching Procedure) show that the heavy metals are securely fixed in the matrix, and any trace elements that manage to leach out remain well below hazardous regulatory limits. [12, 14]

Limitations and Challenges

- **Replacement Limits:** WHA cannot completely replace cement. Exceeding a **15% to 20% replacement level** reduces concrete workability and decreases structural strength due to high water demand and altering the chemical balance. [1, 3, 15]
- **Processing Costs:** Water hyacinths consist of roughly 90% water. Harvesting, drying, and incinerating them in controlled ovens at specific temperatures (usually around 600°C) to get high-quality amorphous silica requires a multi-step setup. [3, 5, 16, 17, 18]
- **Intended Use:** Because of potential trace leaching over long periods, WHA-infused concrete is highly recommended for non-structural, low-risk applications first—such as load-bearing masonry blocks, pavements, or decorative structures—rather than drinking

water tanks. [19, 20]

[1] https://onlinelibrary.wiley.com
[2] https://www.researchgate.net
[3] https://europepmc.org
[4] https://link.springer.com
[5] https://link.springer.com
[6] https://www.mdpi.com
[7] https://www.researchgate.net
[8] https://link.springer.com
[9] https://www.preprints.org
[10] https://www.savethewater.org
[11] https://www.researchgate.net
[12] https://digital.csic.es
[13] https://www.nature.com
[14] https://www.tandfonline.com
[15] https://www.researchgate.net
[16] https://patents.google.com
[17] https://link.springer.com
[18] https://www.irjet.net
[19] https://www.academia.edu
[20] https://www.sciencedirect.com